On Modal Logic of Full Products of Neighborhood Frames

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Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s

$$
\begin{tikzcd}
x \arrow[d, bend left, "F_1"] \\
y \arrow[u, bend left]
\end{tikzcd}
\qquad
\begin{tikzcd}
(x,u) \arrow[r, bend left] \arrow[d, bend left] & (x,v) \arrow[l, bend left, "R_2"] \arrow[d, bend left, "R_1"] \\
(y,u) \arrow[r, bend left] \arrow[u, bend left, "R_1"] & (y,v) \arrow[l, bend left, "R_2"] \arrow[u, bend left]
\end{tikzcd}
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\qquad F_1 \times F_2
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- $\Box_1\Box_2 p \leftrightarrow \Box_1\Box_2 p$ (com) and $\Diamond_1\Box_2 p \to \Box_2\Diamond_1 p$ (chr) are valid

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- $\Box_1\Box_2 p \leftrightarrow \Box_1\Box_2 p$ (com) and $\Diamond_1\Box_2 p \to \Box_2\Diamond_1 p$ (chr) are valid
- $S4 \times S4 = S4 \otimes S4 +$ (com) $+$ (chr)

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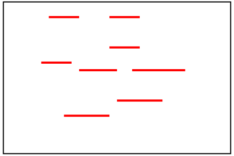

Figure: Element of horizontal topology

Motivation

- (com) and (chr) fails on products of topological frames
- $S4 \times_t S4 = S4 \otimes S4$
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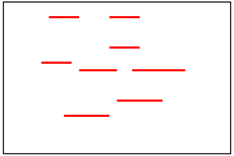

Figure: Element of horizontal topology

Contribution

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-
-
- $S4 \times_t S4 = S4 \otimes S4$
- $S4 \times_t^+ S4 = S4 \otimes S4 \otimes S4 + \Box p \to \Box_1 p \wedge \Box_2 p$

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

Neighbourhood frame

Definition

Let X be a non-empty set and $k \geq 1$. A k-neighborhood frame (n-frame) is a pair (X, τ), where where $\tau : X \to 2^{2^X}$ assigns to every $x \in X$ a *filter* $\tau(x)$ on X, i.e.

- $\tau(x) \neq \emptyset$
- if $U \in \tau(x)$ and $U \subseteq V$ then $V \in \tau(x)$
- if $U, V \in \tau(x)$ then $U \cap V \in \tau(x)$

Full Product of Frames

Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathsf{X} \times \mathsf{Y}, \tau_1', \tau_2', \tau)$ defined as follows :

$$\begin{aligned}
\tau_1'(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\}, \\
\tau_2'(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\}, \\
\tau(x, y) &= \{U \subseteq \mathsf{X} \times \mathsf{Y} \mid \exists V \in \tau_1(x)\ \exists W \in \tau_2(y) : V \times W \subseteq U\}
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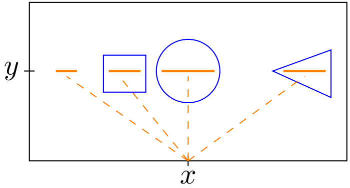

Multimodal Logics

Definition

Let L_1 and L_2 be two unimodal logics. Then the *fusion* ($\otimes$) of these logics is

$$L_1 \otimes L_2 = K_2 + L_1^{\Box \to \Box_1} + L_2^{\Box \to \Box_2}$$

Furthermore the full product ($\times_n^+$) is defined as

$$L_1 \times_n^+ L_2 = Log\{\mathcal{X} \times_n^+ \mathcal{Y} \mid \mathcal{X} \vDash L_1 \text{ and } \mathcal{Y} \vDash L_2\}$$

Main Theorem

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

In the following we will sketch the proof for $\Lambda = T$.

We denote $T \otimes T \otimes T + \Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$ as T_3m

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T^{r,m}_{\omega,\omega,\omega}$) with $Log(T^{r,m}_{\omega,\omega,\omega}) = T_3m$

$$T \otimes T \otimes T + (\Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3 m$$

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- construct a neighbourhood product $N^r_\omega \times^+_n N^r_\omega$ with $Log(N^r_\omega) = T$ s.t. there is surjective bounded morphism from $N^r_\omega \times^+_n N^r_\omega$ to $N(T^{r,m}_{\omega,\omega,\omega})$

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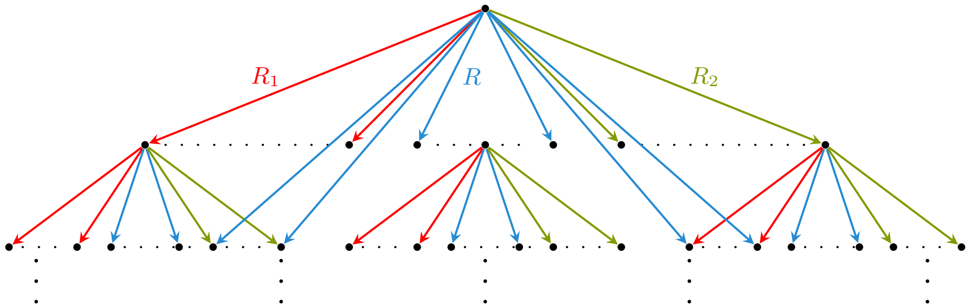
- Keynote: For any Kripke frame $F = (W, R_1, ...R_n)$, we can construct a neighborhood frame $N(F)$, i.e. for any $w \in W$: $\tau_i(w) = \{U \mid R_i(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(N(F))$

$$T \otimes T \otimes T + (\Box p \to \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

$\mathbf{Log}(T^{r,m}_{\omega,\omega,\omega}) = T_3m$



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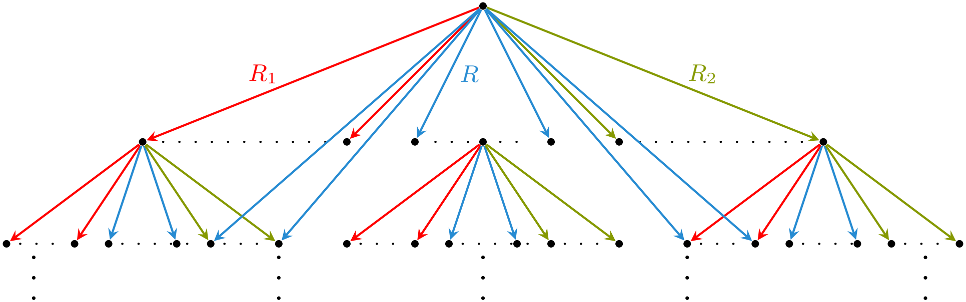

$$T \otimes T \otimes T + \Box p \to \Box_1\Box_2 p \wedge \Box_2\Box_1 p = T_3m$$

- Note: $\Box p \to \Box_1 p \wedge \Box_2 p$ is derivable
- $\mathrm{Log}(T^{r,m}_{\omega,\omega,\omega}) \subseteq T_3m$ by standard techniques

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0 a_1 a_2 ... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N}\ \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences.

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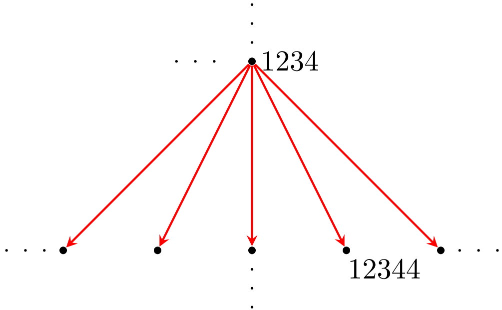
is the set of pseudo infinite sequences. Let $\alpha = a_0 a_1 a_2 ... \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \textit{forg}(\alpha) R \textit{forg}(\beta) \text{ and } \alpha \mid_m = \beta \mid_m, \text{with } m = max(k, st(\alpha))\}$$

where *forg*(α) deletes all $\star$ and $st(\alpha) = min\{N \mid \forall k \geq N : a_k = \star\}$

Example of $U_k(\alpha)$



Let $\alpha = \star\star 12 \star 3 \star 4 \star\star^{\omega} \in X$. Then $st(\alpha) = 9$. Let's consider $U_9(\alpha)$. Then

$$\beta = \star\star 12 \star 3 \star 4 \star 4 \star\star^{\omega} \in U_9(\alpha)$$

Definition of the neighborhood frame N^r_w

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Definition of the neighborhood frame N_w^r

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We define $N_\omega^r = (X, \tau)$ where $\tau : X \to 2^{2^X}$ with

$$\tau(\alpha) = \{U \subseteq X \mid \exists k \in \mathbb{N} : U_k(\alpha) \subseteq U\}$$

Note: τ is a filter and there is no smallest neighborhood

Surjective bounded morphism

Let $N^r_\omega \times^+_n N^r_\omega = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T^{r,m}_{\omega,\omega,\omega}) = ((\mathbb{N} \times \{0, 1, 2\})^*, \sigma_1, \sigma_2, \sigma)$

Surjective bounded morphism

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega,\omega,\omega}^{r,m}) = ((\mathbb{N} \times \{0,1,2\})^*, \sigma_1, \sigma_2, \sigma)$

Theorem

There is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \to N(T_{\omega,\omega,\omega}^{r,m})$.

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$$\begin{array}{llllll} \alpha & = 3 & \star & 2 & \star & \star^\omega \\ \beta & = \star & 1 & 7 & \star & \star^\omega \\ \hline \end{array}$$

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\alpha & = 3 & \star & 2 & \star & \star^\omega \\
\beta & = \star & 1 & 7 & \star & \star^\omega \\
\hline
g'(\alpha,\beta) & = (3,1) & (1,2) & (h(2,7),0) & \star & \star^\omega
\end{array}$$

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Then $g(\alpha, \beta) = (3,1)(1,2)(h(2,7),0)$

Surjective bounded morphism

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Theorem

There is a surjective bounded morphism $g : N^r_\omega \times^+_n N^r_\omega \to N(T^{r,m}_{\omega,\omega,\omega})$.

- surjectivity and
- $R(g(\alpha,\beta)) \subseteq g(U_k(\alpha) \times U_j(\beta)) \subseteq g(U)$ for some $k, j \in \mathbb{N}$

2nd cond: for all $1 \leq i \leq k$, for all $w \in W$ and all $U \in \tau_i(w)$, there is a $V \in \sigma_i(g(w))$ with $V = g(U)$ Note:
$\tau(x,y) = \{U \mid \exists k, j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}, \sigma(x) = \{U \mid R(x) \subseteq U\}$.

Surjective bounded morphism

$g(\alpha, \beta) = (3,2)(1,0)h((7,2),0) R \, (3,2)(1,0)h((7,2),0)(h(6,7),0)$

$$\begin{array}{cccccccc} \alpha' & = \star & 1 & 2 & \star & 6 & \star^\omega \\ \beta' & = 3 & \star & 7 & \star & 7 & \star^\omega \\ \hline g'(\alpha,\beta) & = (3,2) & (1,0) & (h(2,7),0) & \star & (h(6,7),0) & \star^\omega \end{array}$$

Claim: $R(g(\alpha, \beta)) \subseteq g(U_k(\alpha) \times U_j(\beta)) \subseteq g(U)$ for some $k, j \in \mathbb{N}$

Surjective bounded morphism

Let $N^r_\omega \times^+_n N^r_\omega = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T^{r,m}_{\omega,\omega,\omega}) = ((\mathbb{N} \times \{0,1,2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$

Theorem

be two There is a surjective bounded morphism $g : N^r_\omega \times^+_n N^r_\omega \to N(T^{r,m}_{\omega,\omega,\omega})$. Note: $Log(N^r_w) = T$ and hence $N^r_w \times^+_n N^r_w \in T \times^+_n T$

Pick $m > max\{st(\alpha), st(\beta)\}$ and set $U := U_m(\alpha) \times U_m(\beta)$. Then

$$g(U) = g(U_m(\alpha) \times U_m(\beta)) \subseteq R(g(\alpha,\beta)) \subseteq V$$

Note: $\tau(x,y) = \{U \mid \exists k,j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}, \sigma(x) = \{U \mid R(x) \subseteq U\}$
3rd cond: for all $1 \leq i \leq k$, for all $w \in W$ and all $V \in \sigma_i(f(w))$, there exists $U \in \tau_i(w)$ such that $g(U) \subseteq V$

Conlusion

- $Log(T^{r,m}_{\omega,\omega,\omega}) = T_3 m$

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- $Log(T^{r,m}_{\omega,\omega,\omega}) \supseteq Log(N^w_\omega \times^+_n N^w_\omega) \supseteq T \times^+_n T$

Conlusion

- $Log(T^{r,m}_{\omega,\omega,\omega}) = T_3m$
- $Log(T^{r,m}_{\omega,\omega,\omega}) \supseteq Log(N^w_\omega \times^+_n N^w_\omega) \supseteq T \times^+_n T$
- $T3m \supseteq T \times^+_n T$

Future works